

Kummer Row-Balance and the Chebotarev–Matroid Link

Jared Wilder

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Abstract

Paper 9 reduces Kummer collision decay to finite row-balance or angular-energy lower bounds for the Kummer log matrix. This note identifies exactly where those lower bounds come from. Pure finite rank data cannot force row-balance: high-rank matrices may still have all rows concentrated in a hyperplane. For actual Kummer rows, however, fixed- ℓ Chebotarev equidistribution supplies row-balance unconditionally for multiplicatively independent bases. This proves exponential collision decay in the fixed- ℓ , density-floor regime. The high- ℓ regime is separated as a conditional analytic seam: it requires a uniform hyperplane or angular non-concentration hypothesis and is not proved here.

1 Finite row-balance cannot be rank-only

Let $X = (x_q)_{q \in \mathcal{Q}}$ be a matrix with rows $x_q \in \mathbb{F}_\ell^r$. For $A \in \mathbb{F}_\ell^r \setminus \{0\}$, define the hyperplane

$$H_A = \{x \in \mathbb{F}_\ell^r : A \cdot x = 0\}$$

and the cut size

$$d_A(X) = \#\{q \in \mathcal{Q} : x_q \notin H_A\}.$$

Put

$$d(X) = \min_{A \neq 0} d_A(X).$$

Proposition 1 (No rank-only lower bound). *There is no function $f(r, |\mathcal{Q}|) \gg |\mathcal{Q}|$ such that every rank- r row matrix $X \in \mathbb{F}_\ell^{|\mathcal{Q}| \times r}$ satisfies $d(X) \geq f(r, |\mathcal{Q}|)$.*

Proof. Fix a basis vector $e_1 \in \mathbb{F}_\ell^r$. Put $N - r$ rows equal to e_1 , and add $e_2, \dots, e_r$ once each. The resulting matrix has rank r . Let $A \neq 0$ satisfy $A \cdot e_1 = 0$ and $A \cdot e_i \neq 0$ for at least one $i \geq 2$; such an A exists for $r \geq 2$. Then only the $O(r)$ exceptional basis rows are cut by A , so $d(X) \leq r - 1$, independently of $N = |\mathcal{Q}|$. Thus no linear lower bound follows from rank alone. $\square$

Remark 1. *This is the finite myopia guardrail. Paper 9 supplies a perfect finite machine, but the machine needs distributional fuel. In arithmetic applications that fuel is Kummer-log non-concentration.*

2 Kummer logs as Frobenius coordinates

Fix a prime ℓ . Let

$$\mathbf{a} = (a_1, \dots, a_r) \in (\mathbb{Q}^*)^r$$

be multiplicatively independent modulo ℓ -th powers in $\mathbb{Q}(\zeta_\ell)^*$. Put

$$K_\ell = \mathbb{Q}(\zeta_\ell, a_1^{1/\ell}, \dots, a_r^{1/\ell}).$$

Then

$$\text{Gal}(K_\ell/\mathbb{Q}(\zeta_\ell)) \cong \mathbb{F}_\ell^r.$$

For primes $q \nmid \ell a_1 \cdots a_r$ with $q \equiv 1 \pmod{\ell}$, the Frobenius class over $\mathbb{Q}(\zeta_\ell)$ determines a Kummer log row

$$x_q \in \mathbb{F}_\ell^r.$$

For $A = (A_1, \dots, A_r) \neq 0$, the condition

$$A \cdot x_q = 0$$

is equivalent to

$$a_1^{A_1} \cdots a_r^{A_r} \in (\mathbb{F}_q^*)^\ell.$$

3 Fixed- ℓ row-balance

Let

$$\mathcal{Q}_\ell(X) = \{q \leq X : q \equiv 1 \pmod{\ell}, q \nmid \ell a_1 \cdots a_r\}.$$

Theorem 1 (Fixed- ℓ Kummer row-balance). *Assume ℓ , r , and $\mathbf{a}$ are fixed, and that the classes of $a_1, \dots, a_r$ are independent in $\mathbb{Q}(\zeta_\ell)^*/(\mathbb{Q}(\zeta_\ell)^*)^\ell$. Then, as $X \rightarrow \infty$, for every $A \in \mathbb{F}_\ell^r \setminus \{0\}$,*

$$\#\{q \in \mathcal{Q}_\ell(X) : A \cdot x_q = 0\} = \frac{1}{\ell} |\mathcal{Q}_\ell(X)| + o(|\mathcal{Q}_\ell(X)|),$$

and therefore

$$d_A(X) = \#\{q \in \mathcal{Q}_\ell(X) : A \cdot x_q \neq 0\} = \left(1 - \frac{1}{\ell}\right) |\mathcal{Q}_\ell(X)| + o(|\mathcal{Q}_\ell(X)|).$$

Since the set of nonzero A 's is finite for fixed ℓ, r , the same estimate holds uniformly in $A \neq 0$.

Proof. The condition $A \cdot x_q = 0$ cuts out a hyperplane in $\text{Gal}(K_\ell/\mathbb{Q}(\zeta_\ell)) \cong \mathbb{F}_\ell^r$, hence a proportion $1/\ell$ of the relative Galois group. Chebotarev's density theorem over the fixed extension $K_\ell/\mathbb{Q}(\zeta_\ell)$, restricted to rational primes $q \equiv 1 \pmod{\ell}$, gives the stated asymptotic. There are only $\ell^r - 1$ nonzero dual directions, so the error remains $o(|\mathcal{Q}_\ell(X)|)$ uniformly after taking the maximum over $A \neq 0$. $\square$

4 Angular energy and collision decay

For $q \in \mathcal{Q}_\ell(X)$, let

$$p_q = \frac{|\Gamma_q|}{q-1}.$$

Define

$$\mathcal{E}(A) = \sum_{q \in \mathcal{Q}_\ell(X)} 2p_q(1-p_q) \left(1 - \cos \frac{2\pi(A \cdot x_q)}{\ell}\right).$$

Theorem 2 (Fixed- ℓ angular-energy floor). *Assume the hypotheses of Theorem 1. If there is a constant $0 < \alpha \leq 1/2$ such that*

$$\alpha \leq p_q \leq 1 - \alpha \quad (q \in \mathcal{Q}_\ell(X)),$$

then

$$\min_{A \neq 0} \mathcal{E}(A) \geq 2\alpha(1 - \alpha)|\mathcal{Q}_\ell(X)|(1 - o(1)).$$

Proof. For fixed $A \neq 0$, Chebotarev gives equidistribution of $A \cdot x_q$ in $\mathbb{F}_\ell$. Hence

$$\sum_{q \in \mathcal{Q}_\ell(X)} \left(1 - \cos \frac{2\pi(A \cdot x_q)}{\ell} \right) = |\mathcal{Q}_\ell(X)|(1 - o(1)),$$

because

$$\frac{1}{\ell} \sum_{t \in \mathbb{F}_\ell} \cos \frac{2\pi t}{\ell} = 0.$$

The density-floor assumption gives $p_q(1 - p_q) \geq \alpha(1 - \alpha)$. Since the nonzero set of A 's is finite, the estimate is uniform in $A \neq 0$. $\square$

Corollary 1 (Fixed- ℓ exponential collision decay). *Under the hypotheses of Theorem 2, the normalized collision excess from Paper 8 satisfies*

$$\frac{\mathcal{C}_\ell(N)}{\varphi(P)^2} \leq (1 - \ell^{-r}) \exp(-2\alpha(1 - \alpha)|\mathcal{Q}_\ell(X)|(1 - o(1))).$$

Proof. Apply the Paper 9 angular-energy bound

$$\frac{\mathcal{C}_\ell(N)}{\varphi(P)^2} \leq (1 - \ell^{-r})e^{-\mathcal{E}_*}, \quad \mathcal{E}_* = \min_{A \neq 0} \mathcal{E}(A),$$

and then substitute Theorem 2. $\square$

5 The high- ℓ seam

The previous section is fixed- ℓ . It does not prove the high- ℓ non-concentration needed in the analytic seam of the EG203 program. As ℓ grows, the degree and discriminant of K_ℓ grow, and fixed-field Chebotarev error terms no longer supply uniform row-balance in the range $\ell \leq (\log X)^B$.

Hypothesis 1 (Weighted angular non-concentration). *There is a constant $\eta > 0$ such that, uniformly for all nonzero $A \in \mathbb{F}_\ell^r$,*

$$\sum_{q \in \mathcal{Q}} p_q(1 - p_q) \left(1 - \cos \frac{2\pi(A \cdot x_q)}{\ell} \right) \geq \eta|\mathcal{Q}|.$$

Corollary 2 (Conditional high- ℓ collision decay). *If Hypothesis 1 holds for a finite prime set $\mathcal{Q}$, then*

$$\frac{\mathcal{C}_\ell(N)}{\varphi(P)^2} \leq (1 - \ell^{-r})e^{-2\eta|\mathcal{Q}|}.$$

Proof. The hypothesis gives $\mathcal{E}_* \geq 2\eta|\mathcal{Q}|$. The Paper 9 angular-energy bound gives the result. $\square$

Remark 2 (Relation to PHNC). *For $r = 2$ and $\mathbf{a} = (2, 3)$, the condition $A \cdot x_q = 0$ is exactly*

$$2^{A_1} 3^{A_2} \in (\mathbb{F}_q^*)^\ell.$$

Thus hyperplane row-balance is the finite-matrix form of prime Hecke non-concentration. The angular hypothesis above is stronger for odd ℓ , because it controls not only whether $A \cdot x_q$ vanishes but also whether nonzero values cluster near small angles.

6 Dependency status

- [Proposition 1](#) is finite linear algebra.
- [Theorems 1](#) and [2](#) and [Corollary 1](#) are unconditional for fixed ℓ , conditional only on the stated multiplicative independence and density-floor assumptions.
- [Corollary 2](#) is explicitly conditional on weighted angular non-concentration.
- No high- ℓ PHNC theorem, Kummer-BV theorem, GRH theorem, or Erdős–Graham #203 conclusion is asserted.